
Will LLMs Scaling Hit the Wall? Breaking Barriers via Distributed Resources on Massive Edge Devices

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Abstract

The remarkable success of foundation models has been driven by scaling laws, demonstrating that model performance improves predictably with increased *training data* and *model size*. However, this scaling trajectory faces two critical challenges: the exhaustion of high-quality public data, and the prohibitive computational power required for larger models, which have been monopolized by tech giants. These two bottlenecks pose significant obstacles to the further development of AI. In this position paper, we argue that leveraging massive distributed edge devices can break through these barriers. We reveal the vast untapped potential of data and computational resources on massive edge devices, and review recent technical advancements in distributed/federated learning that make this new paradigm viable. Our analysis suggests that by collaborating on edge devices, everyone can participate in training large language models with small edge devices. This paradigm shift towards distributed training on edge has the potential to democratize AI development and foster a more inclusive AI community. The project page is available at <https://tao-shen.github.io/Distributed-LLM-Edges/>

1 Introduction

Scaling laws [1, 2] have been fundamental to the remarkable success of foundation models, demonstrating a predictable relationship between performance and the expansion of model parameters and training data. These laws have guided the development of increasingly powerful models, from BERT [3] to GPT-4 [4], showing that performance improvements can be achieved through systematic scaling of both model size and training data [5, 6]. However, the continued application of these scaling laws requires ever-increasing amounts of data and computational resources, pushing the boundaries of what is currently feasible [7].

Public data has been the primary *fuel* driving AI development forward. This field has witnessed an exponential growth in data requirements, from the early success of MNIST [8] with its 70,000 handwritten digits to ImageNet’s revolutionary impact with 14 million labeled images [9]. This trajectory has continued with modern large language models (LLMs) like GPT [4], LLaMA [10], and DeepSeek [11] series, which are trained on trillions of tokens. Recent evidence that LLaMA 3.1’s smallest model (8B) [12] trained on 15 trillion tokens, outperforms LLaMA 2’s largest model (70B) [13] trained on 2 trillion tokens (despite being $10\times$ smaller in model size, the $7\times$ increase in training data leads to superior performance), demonstrates the paramount importance of data scaling [14, 15]. However, we are witnessing a concerning trend of data exhaustion, where high-quality public data sources are becoming exhausted [16, 17]. [18] argues that human-generated public text data cannot sustain scaling beyond this decade. While recent efforts advocate for training larger models with synthetic data [19], AI-generated content may fail to yield performance improvements [20], also risks

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polluting public data sources [21]. Moreover, stricter data privacy regulations like GDPR [22] have made data collection increasingly difficult and expensive. This looming data scarcity suggests that scaling laws may hit a wall [23], potentially impeding further AI advancement.

Computational resources has been the primary *engine* powering AI development. Throughout AI history, major breakthroughs have been closely tied to advances in computing power, from early models requiring single CPUs (with peak performance of 1-2 GFLOPS) to modern GPU clusters. The computational demands have grown exponentially - from BERT-Large’s training requiring 64 TPU v3 chips (providing 420 TFLOPS) [3] to GPT-3’s training on 10,000 V100 GPUs (reaching 28,000 TFLOPS) [5], while training GPT-4 reportedly required over 25,000 NVIDIA A100 GPUs (delivering a staggering 400,000 TFLOPS) [4]. More recent models like Grok 3 push these requirements even further [24]. However, we are approaching physical limits in single-chip performance as Moore’s Law slows down [25]. While massive computing clusters can compensate for individual chip limitations, maintaining such infrastructure incurs astronomical costs - estimated at over \$100M for training GPT-4 [26] - and poses significant environmental concerns due to their enormous energy consumption, with each training run emitting as much CO₂ as 500 cars driven for a year [27]. Moreover, this level of computing power has become concentrated among a few tech giants, creating a monopolistic landscape that effectively excludes smaller companies and academic institutions from participating in foundational AI research [28]. This centralization of computing resources presents a significant barrier to innovation and democratization in AI development [29].

In this paper, we propose that leveraging massive distributed edge devices offers a promising solution to overcome both data and computing barriers in AI development. Our analysis (using smartphone as an example) reveals two compelling opportunities: First, edge data generated from smartphones for past 5 years are projected to reach 33.1 EB, offering fresh, diverse, and contextually rich training samples. Second, the collective computing power of edge devices - with smartphones delivering 9,278 EFLOPS for past 5 years - demonstrates the feasibility of distributed model training, as training state-of-the-art models like DeepSeek-v3 would require only about 60,723 users with edge devices working (*ideally*) in parallel to match its current training setup. Based on these insights, **we argue that leveraging these massive distributed edge devices can break barriers of data and computing wall, and everyone can participate in training large models with small edge devices.** To support this position, we first analyze the critical challenges of large language models, examining both data bottlenecks (§2.1) and computational monopolization (§2.2). We then explore the hidden potential of massive edge devices, investigating their vast untapped distributed data resources (§3.1) and computational capabilities (§3.2). Building on these insights, we investigate technical approaches for overcoming large model challenges through distributed computing architectures (§4): small language models at edges (§4.1), collaborative inference (§4.2), and collaborative training (§4.4). We then identify two critical open challenges: heterogeneous device model fusion and heterogeneous device compute sharing (§5). Finally, we discuss the societal impact like AI democratization, incentive mechanisms, and environmental benefits of this paradigm shift (§A).

2 Scaling at Risk: Challenges of Data and Computing Power

2.1 The Ceiling of Public Data

Public data for pretraining is exhausting. The rapid advancement of large language models has created an insatiable appetite for training data. Scaling laws establish that model performance improves predictably with data quantity—a relationship that demands exponentially growing datasets [2]. A canonical example is GPT-3, trained on 300 billion tokens spanning books, web content, and programming code [5]. Current projections suggest dataset sizes grow at 0.38 orders of magnitude ($2.4\times$) annually [30], implying models will require three orders of magnitude more data within a decade.

Despite the internet’s vast textual resources, the total stock of high-quality human-generated text remains bounded. Recent estimates place this limit at approximately 4×10^{14} tokens [30]. [31] argues that current consumption patterns suggest exhaustion of public text data by 2028, potentially accelerated to 2026 through excessive data reuse during training (a practice termed overtraining). Therefore, the finite nature of publicly available human-generated text data is expected to become a major bottleneck for LLM scaling within the next decade. Despite the current large scale of public data, the risk of data exhaustion is rapidly approaching as data demand continues to grow [32].

Synthetic data has potential but faces challenges. Faced with the threat of data exhaustion, researchers have proposed various solutions, among which synthetic data generation is considered

one of the most promising approaches. By leveraging LLMs to produce their own training data, researchers envision self-sustaining data ecosystems. Early successes in constrained domains like mathematics and code generation, where automated verification ensures quality, demonstrate potential [33]. Recent work [19] demonstrated that diverse synthetic data enhances the performance of LLMs during both pre-training and fine-tuning.

The adoption of synthetic data faces three fundamental challenges. First, *model collapse* occurs when models iteratively train on their own outputs, causing gradual divergence from original data distributions. This recursive process amplifies biases and reduces output diversity, ultimately degrading model performance across generations [34, 35]. Second, *synthetic data quality* remains inherently unverifiable in open-domain contexts. While formal domains like mathematics allow algorithmic validation, natural language lacks objective evaluation standards. The absence of ground-truth verification creates self-referential quality assessments, compromising reliability [36, 20]. Finally, synthetic data struggles to replicate human *linguistic diversity*. Current methods disproportionately replicate dominant language patterns while underrepresenting cultural nuances and low-frequency expressions. This homogeneity limits their utility for training robust general-purpose models [37].

These persistent challenges underscore that synthetic data alone cannot sustainably address the looming data scarcity crisis, compelling the research community to seek complementary strategies that transcend conventional data acquisition paradigms.

2.2 The Monopoly of Computing Resources

A few AI giants dominate the computing power. The AI computing landscape is dominated by a few major tech giants like OpenAI, Google, Microsoft, and Meta, which control powerful hardware such as GPUs and TPUs. This monopolization creates a significant barrier for smaller AI startups and research institutions, who struggle to access such advanced resources. Additionally, these companies control proprietary AI models, datasets, and software frameworks that require immense computing power, further widening the gap between the giants and smaller players. As a result, high-performance computing resources remain increasingly inaccessible to anyone outside these dominant entities.

Computational demand is growing exponentially. As large-scale AI models like GPT-4 [4], Llama 3 [12], and DeepSeek-V3 [11] surpass the trillion-parameter scale, the global AI landscape faces severe computational efficiency challenges. As shown in Figure 1, since the deep learning revolution in 2010, AI training demands have grown at a super-exponential rate of $3.9\times$ per year—an acceleration that intensified with the adoption of the transformer architecture as the industry standard [39]. With the advent of the era of large language models in 2022, the demand for computing power has surged even further, reaching an unprecedented growth rate of $12.8\times$ per year. This marks a transformative shift in AI computation, where the need for computing power is expanding at an unprecedented pace, pushing the limits of existing hardware and infrastructure.

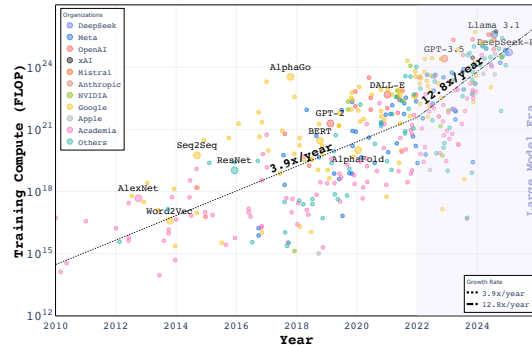


Figure 1: Trend of Computational Demand for Model Training. (Data source: [38]).

Moore’s Law is slowing down. Moore’s Law, which has driven the growth in computing power for decades, is slowing down as we approach the physical limits of silicon-based chip technology [40]. The difficulty in shrinking transistors has led to diminishing returns in computational performance. As a result, the AI industry is relying more on specialized hardware like GPUs, TPUs, and custom chips to meet growing demands. However, this shift has made high-performance hardware even more expensive and exclusive, further intensifying the gap between organizations with the resources to develop advanced AI models and those without.

Infrastructure capacity is a constraint. The rapid expansion of AI model scales and the surge in computational demand are facing dual constraints in global computing infrastructure. On one hand, bottlenecks in advanced semiconductor manufacturing severely limit the expansion rate of AI data centers. The foundry capacity for wafers at 5nm and below—such as those produced by TSMC—has already been fully booked by leading technology companies until 2026 [41]. Moreover,

the construction of new wafer fabs involves long lead times and is further constrained by the global supply chain shortages of critical equipment, such as lithography machines. On the other hand, the exponential increase in chip deployment within individual AI clusters is putting immense pressure on the already limited semiconductor manufacturing capacity, pushing the industry toward its production ceiling [42]. These factors have significantly hindered the continuous expansion of computing power, making it increasingly difficult to scale AI infrastructure sustainably.

3 Scaling Beyond Limits: Opportunities from Edge Devices

3.1 Massive Data from Edges

As discussed in § 2.1, edge data represents a crucial alternative to synthetic data in addressing the challenge of data exhaustion. Edge data refers to the data generated by edge devices at or near the source of data generation, which typically remains private and localized rather than being publicly accessible. Edge devices encompass a wide range of equipment including Internet of Things (IoT) sensors, smartphones, wearables, industrial controllers, and other smart devices that process data at the network edge. Data generated at edges offers unique advantages in both data volume and data quality.

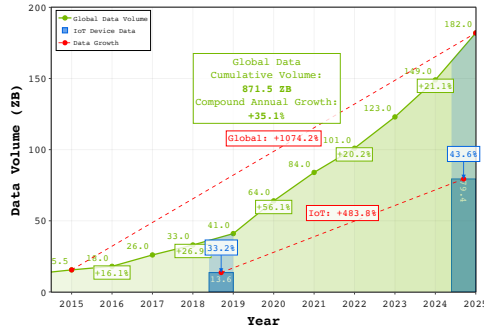


Figure 2: Global data volume from 2014 to 2025 and IoT device data volume in 2015 and 2025. (Data sources: Global data volume from [43]; IoT device data volume from [44].)

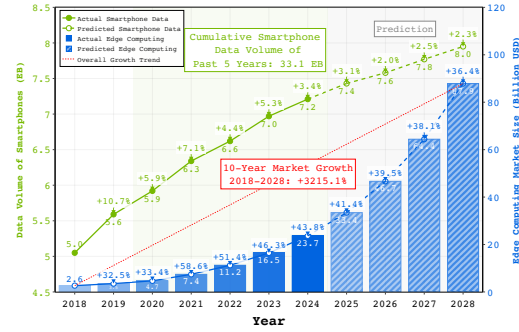


Figure 3: Smartphone data volume with edge computing market size (right) from 2018 to 2028. (Data sources: Edge computing market [45]; Smartphone data volume from [46].)

Edge-generated data is explosively growing. According to the statistical data from [43, 44] (as illustrated in Figure 2), the global data volume is projected to reach 182 ZB by 2025 [43], where the data generated by IoT devices is anticipated to increase from 13.6 ZB in 2019 to 79.4 ZB in 2025 [44], elevating its share of the global data volume from 33.2% to 43.6%, showing a particularly pronounced growth in edge-generated data. Over the period from 2015 to 2025, the global data volume exhibited a compound annual growth rate (CAGR) of 35.1%, resulting in an overall increase of 1074.2% and a cumulative total of 871.5 ZB. IoT device data experienced a growth of 483.8% from 2019 to 2025. This trend underscores the increasingly central role of edge-generated data in the global data ecosystem. Beyond IoT devices, smartphones, as a critical source of edge-side data, are also contributing to the steady rise in data volume. As depicted in Figure 3, the estimated smartphone data volume is projected to grow from 5 EB in 2018 to 8 EB by 2028³. This exponential growth is closely aligned with the rapid expansion of the edge computing market, which is forecasted to surge from \$5.5 billion in 2019 to \$87.9 billion by 2028, representing a remarkable growth rate of 3215.1%. The burgeoning edge computing market has further catalyzed the generation and processing of edge-side data, reinforcing its significance in the broader data landscape⁴.

Edge-generated data has distinctive advantages. Beyond its impressive quantity, edge data possesses several distinctive characteristics that make it particularly valuable for model training. First, edge data provides enhanced *privacy* characteristics. Since edge data typically remains local to devices and does not need to be centrally stored, it allows for more privacy-preserving approaches

³These numbers are estimated based on an average user data generation of 1 GB per device. For detailed estimation methodology, refer to Appendix C.

⁴[47, 48] provide a more comprehensive overview of the global data volume, but we cannot access the statistics data. We appreciate any suggestions for better statistical data sources.

to data utilization. This local-first nature enables compliance with increasingly strict data privacy regulations while still allowing the data to contribute to model training. Second, edge data exhibits superior *diversity* across multiple dimensions. It encompasses a wide variety of data types from IoT devices, mobile interactions, and personal devices, covering different domains, languages, and user behaviors. This natural diversity provides richer training signals compared to curated public datasets [49]. Third, edge data demonstrates strong *real-time* capability. Unlike public datasets, which are often updated infrequently, edge devices continuously generate fresh data with low latency [50], offering more up-to-date and relevant training samples. Despite these advantages, edge-generated data can present challenges such as low signal-to-noise ratios, unclear, or potentially harmful content. However, recent advancements in data quality assessment methods (e.g., data prospector [51]), have emerged to identify and filter high-quality data from edge sources, ensuring that only the most reliable and valuable data is selected for model training.

In conclusion, edge data with its explosive growth and distinctive characteristics, is a valuable resource for model pretraining. Its diversity, real-time nature, personalization, and rich context make it an ideal foundation for developing robust and adaptable large-scale models, enhancing their ability to serve real-world applications effectively.

Insight: The smartphone data volume of the past 5 years (before 2025) is projected to reach approximately 33.1 EB, with unique advantages in privacy, diversity, and real-time context, demonstrating the massive data potential of edge for AI model training.

3.2 Massive Computing Power from Edges

Edge computing power is growing rapidly. In recent years, edge computing has experienced explosive growth in computing power, driving smart devices to evolve from single-function tools into multimodal perception and decision-making centers. For instance, as shown in Figure 4, flagship smartphones such as the iPhone 16 series, equipped with 3nm process chips, have achieved computing power exceeding 2 TFLOPS [52], enabling local execution of complex AI tasks like real-time image enhancement and multilingual speech translation. Notably, the computing power of an individual smartphone has potentially surpassed that of laptops in the same period. The breakthroughs are even more pronounced in the desktop sector, achieving an annual computing power growth rate of $1.29\times/\text{year}$, surpassing that of smartphones and laptops (both $1.20\times/\text{year}$). These three types of devices form a differentiated growth hierarchy, collectively driving edge computing’s overall computing power to expand at an annual average rate of $1.28\times/\text{year}$. This growth is fueled by three key technological drivers: advanced manufacturing processes (3nm technology increases transistor density by 60% [52]), dedicated architectures (modern smartphone SoCs integrate NPUs for AI acceleration), and scenario-driven innovation (e.g., autonomous driving demands end-to-end latency of less than 100ms [53]).

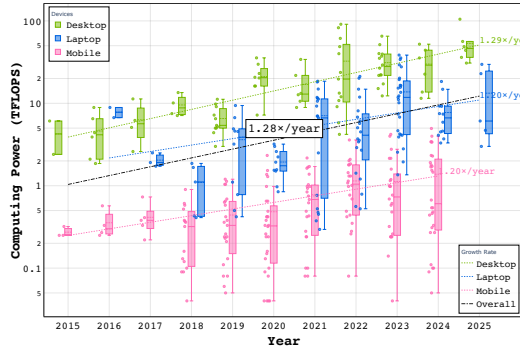


Figure 4: Edge Computing Power Evolution Trend. (Data source: [54]).

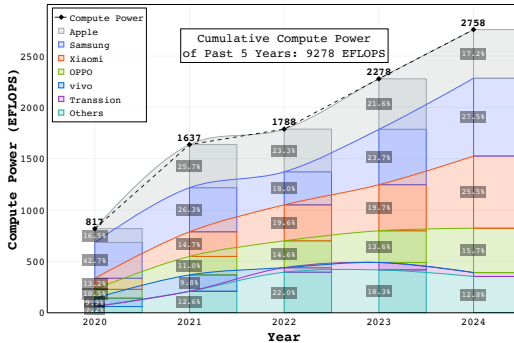


Figure 5: Smartphone Market Share and Computing Power Trends. (Data source: [55]).

Edge computing has potential for LLM training. We analyze the performance of smartphone chips, representing typical edge devices, and estimated their overall computing power. To ensure our estimation is as accurate as possible, we based our calculations on the market share data from [55]. We then estimated the total computing power of newly produced mobile devices by averaging the chip

performance of each brand. From 2020 to 2024, smartphone chip performance has seen significant improvements, with peak computing power increasing from 1.53 TFLOPS to 4.95 TFLOPS, and average computing power rising from 0.48 TFLOPS to 1.38 TFLOPS. Meanwhile, the overall computing power of mobile devices has grown from 817 EFLOPS in 2020 to 2,758 EFLOPS in 2024, and totally 9278 EFLOPS for past 5 years. This trend highlights the rapid expansion of edge computing power, which is not only essential for AI applications but also holds the potential for training complex AI models. For instance, training the DeepSeek-v3 [11] model utilizes 2048 H100 GPUs, each providing a peak FP32 performance of 59.30 TFLOPS, resulting in a total computational capacity of 121,446.4 TFLOPS. If this workload were distributed across edge devices with a peak performance of 2 TFLOPS (e.g., mobile chips like the iPhone 16 series), approximately 60,723 users with edge devices working (*ideally*) in parallel would be required to match the computational capacity.

Insight: The smartphone computing power of the past 5 years (before 2025) is projected to reach approximately 9278 EFLOPS, with individual flagship devices now achieving over 2 TFLOPS performance. The combined parallel computing power of approximately 30 iPhone devices (with A18 chips) can match the computational capacity of a professional AI training GPU (H100 with 59.30 TFLOPS).

However, current smartphone chips are primarily optimized only for inference efficiency rather than training capabilities. *We advocate for and predict a future trajectory of edge computing where smartphone chip designs will increasingly prioritize and optimize on-device model training capabilities.* As computational power grows and distributed algorithms develop, we expect a paradigm shift enabling collaborative model training across networks of edge devices. This evolution positions the edge computing ecosystem as a critical catalyst for democratizing AI development and driving the next wave of innovations in the field.

4 Technical Advancements

4.1 Small Language Models at Edges

The first move of AI to Edge is to deploy small language models (SLMs) to edge devices [56–58]. This trend is driven by the growing demand for AI applications that can run directly on edge devices, motivated by needs for privacy, offline usage, and real-time processing without cloud dependence. However, edge devices have limited memory, computation, and energy resources, requiring more efficient and compact models.

SLMs leverage innovative architectures for efficient edge deployment. The classic Transformer architecture [39] uses self-attention mechanisms for effective sequence modeling but faces quadratic complexity challenges, with models like TinyBERT [59] (14.5M parameters) and ALBERT [60] (12M parameters) demonstrating its effectiveness at small scales. Mamba [61], based on state space models, achieves linear complexity and faster inference by utilizing only the previous hidden state, as demonstrated by Zamba2-2.7B [62] which achieves twice the speed and 27% reduced memory overhead compared to traditional models. Hymba [63] combines both approaches by integrating attention and SSM heads within the same layer for parallel processing, with its 1.5B variant trained on DCLM-Baseline-1.0 and SmoLM-Corpus achieving 11.67 times cache size reduction while outperforming Llama-3.2-3B. The xLSTM architecture [64] modernizes LSTM with exponential gates and matrix memory cells, with models ranging from 125M to 1.3B parameters trained on 300 billion tokens from SlimPajama [65], consistently outperforming comparable RWKV-4 [66], Llama [10], and Mamba models across various tasks in the PALOMA benchmark [67]. These architectural innovations demonstrate the potential for efficient and powerful language models that can run effectively on edge devices.

SLMs can be constructed through diverse methodological approaches. The construction of efficient SLMs relies on a comprehensive suite of techniques, each with specific performance trade-offs. For training SLMs from scratch, optimized MLM approaches [68] with increased masking ratio (25% vs traditional 15%) demonstrate 2-3% performance improvements for models under 3B parameters. When deriving SLMs from existing LLMs, knowledge distillation has proven particularly effective, with response-based distillation [69] reducing model size by 40% while maintaining 95% of the original performance. In architecture optimization, the Mixture of Experts approach [70] enables 65% parameter reduction while potentially improving performance in specific tasks. Domain

specialization has shown remarkable results, particularly in the medical field where 3B parameter models achieve 92% accuracy, outperforming 175B models (89%) [71]. The combination of these techniques yields impressive results - a notable example is a 770M parameter model from [72] that combines distillation, quantization, and domain specialization to achieve 95% of the performance of a 540B model on specific tasks while requiring less than 0.15% of the computational resources. Most successful SLMs achieve their efficiency by combining multiple techniques, typically starting with knowledge distillation, applying compression methods, and finishing with domain-specific fine-tuning. [57] has provided a comprehensive survey of SLMs, and we summarize the architecture innovations and training methods in Appendix E.

Despite remarkable progress in deploying compressed models to edge devices, the current landscape remains largely confined to individual devices operating in isolation, failing to leverage the massive distributed computing power that could be achieved through collaborative training across edge devices. This represents a significant missed opportunity, as the collective computing resources of billions of edge devices worldwide remain untapped, while individual devices struggle with the computational demands of modern AI applications.

4.2 Collaborative Inference at Edges

The emergence of collaborative inference at the edge represents a significant shift in AI infrastructure, enabling more accessible and cost-effective AI solutions compared to traditional data centers which often present barriers in terms of cost, energy consumption, and accessibility. Recent frameworks like Exo [73] enable users to create AI clusters using everyday devices such as phones, tablets, and computers through peer-to-peer architecture, effectively unifying their computational resources.

Several approaches advance this paradigm: Neurosurgeon [74] introduces a lightweight scheduler that partitions DNN computations between devices and datacenters; MoE² [75] optimizes LLM inference under energy and latency constraints; Edgent [76] enables low-latency edge intelligence through adaptive partitioning and early-exit mechanisms; and Galaxy [77] leverages hybrid model parallelism to efficiently execute Transformer models across edge devices. Despite these advances, the current landscape has yet to fully capitalize on the potential of distributed data resources at the edge. The next frontier lies in transforming these devices from mere inference endpoints into active training participants, representing a significant opportunity for distributed AI development.

4.3 Feasibility of On-Device Training

Recent advancements in optimization techniques have significantly reduced the memory and computational requirements for training machine learning models, making on-device training increasingly feasible, even on resource-constrained edge devices. For instance, Lin et al. [78] demonstrated training neural networks on microcontroller units with only 256KB of RAM by employing an algorithm-system co-design framework. Similarly, Cai et al. [79] introduced a tiny transfer learning approach that freezes most parameters and only trains a small subset, allowing effective fine-tuning with minimal memory requirements. Qiu et al. [80] proposed ZeroFL, a framework that relies on highly sparse operations to accelerate on-device training in federated learning settings, enabling efficient model training on edge devices with up to 95% sparsity. Recent work by Sugiura and Matsutani [81] further advanced this field with ElasticZO, which combines zeroth-order and first-order optimization to achieve a better trade-off between accuracy and training cost. Their ElasticZO-INT8 variant achieves integer arithmetic-only training, further reducing memory usage and training time by approximately 1.5x without compromising accuracy. These advancements suggest that on-device training is no longer limited to high-end devices with abundant resources. Even small embedded systems with memory capacities measured in kilobytes rather than gigabytes can participate in model training.

4.4 Collaborative Training at Edges

To harness the potential of vast amounts of data distributed across numerous devices, we envision a future where everyone can participate in training large-scale models. Federated learning emerges as a paradigm for distributed collaborative training that makes this vision possible.

Federated learning [82] is a practical paradigm that enables collaborative model training while preserving data privacy. Instead of collecting raw data from edge devices, which may violate privacy regulations like GDPR [22], this approach distributes the training process across multiple devices. Each device trains on its local data and only shares model updates with the central server. This approach can effectively utilize both computational and data resources available at edges.

Federated LLMs for fine-tuning has emerged as a critical direction in recent research of large language models, addressing the challenges of privacy preservation and resource constraints. FlexLoRA [83] introduces a novel framework that enables efficient fine-tuning of large language models in a federated setting, demonstrating comparable performance to centralized fine-tuning while maintaining data privacy. FedFM [84] tackles the critical challenges of system and statistical heterogeneity in federated learning, proposing adaptive optimization techniques that improve model convergence across diverse client devices. To address the computational constraints of edge devices, FedPETuning [85] employs parameter-efficient fine-tuning techniques, significantly reducing the memory and computation requirements while maintaining model performance. This approach enables even resource-constrained devices to participate in the fine-tuning process. Similarly, [86] bridges the gap between federated learning and foundation models by introducing novel techniques for efficient knowledge transfer and model adaptation in multi-modal heterogeneous federated learning settings. In domain-specific applications, FedMatch [87] demonstrates the effectiveness of federated learning for question-answering tasks, showing that models can be fine-tuned on sensitive domain-specific data while preserving privacy. These advancements are supported by open-source frameworks like Flower [88] and FATE-LLM [89], which provide robust platforms for implementing federated fine-tuning of large language models.

Federated LLMs for pretraining have opened up exciting new possibilities for large language model training. Rather than relying on traditional data center approaches, researchers have developed innovative geographically distributed frameworks that enable collaborative training across many devices. Notably, Prime Intellect [90] has launched the first decentralized training project for a 10 billion parameter model, named INTELLECT-1, which utilizes the OpenDiLoCo framework to significantly reduce communication costs between nodes. This innovative approach allows for dynamic management of computational resources across multiple locations, achieving an impressive 83% overall computational utilization while collaborating with up to 112 H100 GPUs across five countries and three continents. The model not only enhances parameter efficiency by 25 times but also demonstrates robust performance in various benchmark tests. In parallel, the Flower Lab [88] has introduced FlowerLLM, which successfully trained a 1.3 billion parameter large language model (LLM) using novel federated learning methods. Additionally, it has also developed Photon [91], an open-source framework that provides flexible configurations for training models of different sizes, making federated LLM training more accessible to a broader range of participants and computational resources.

These frameworks underscore a shift towards decentralized AI inference and training (summarized in Appendix F), enabling researchers worldwide to contribute to advanced model development without the constraints of centralized resource control, thus paving the way for a more collaborative and inclusive AI landscape.

5 An Open Problem: How to Train LLMs with Small Edge Devices?

A fundamental limitation of traditional federated learning lies in its requirement for each participant to maintain and train a complete model locally. This assumption becomes particularly problematic in the context of large language models, where the computational and memory requirements far exceed the capabilities of most individual participants. For instance, in sensitive domains like healthcare [92], multiple hospitals may wish to collaboratively train a medical language model to leverage their collective data while maintaining data privacy. However, traditional federated learning mandates that each participating hospital possess sufficient computational resources to train the complete model locally. Despite these institutions’ valuable data contributions and their motivation to enhance model capabilities through collaborative training, many hospitals lack the necessary infrastructure to participate effectively. This resource constraint significantly limits the potential for collaborative model training in critical domains where data privacy is paramount but computational resources are unevenly distributed [93]. While some federated learning approaches allow training partial model parameters [94, 95], the enormous disparity between large language models and what edge devices can train—often orders of magnitude smaller due to inherently constrained resources—remains too vast to be effectively bridged by current FL frameworks. Therefore, we need to develop a new federated learning paradigm that enables participants to collaboratively train a large language model even under extremely limited resources (as shown in Figure 6). **It is still an open problem to train large language models with small edge devices.** Therefore, we encourage the research community to develop novel distributed collaborative computing methods in two key directions:

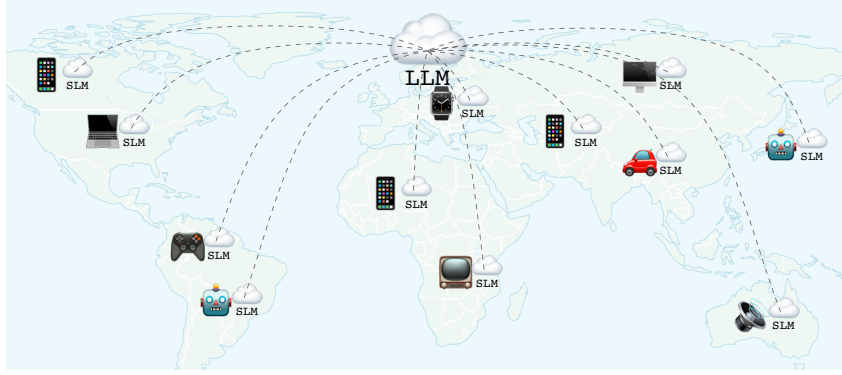


Figure 6: Train Large Language Models with Small Edge Devices

5.1 Heterogeneous Device Model Fusion: from Small to Large

The first direction addresses the fundamental challenge of model size disparity in federated learning. Modern large language models typically contain hundreds of billions of parameters, while edge devices have severely limited computational resources. This creates an enormous scale gap - the large target model may be hundreds or even thousands of times larger than what individual devices can handle. To bridge this gap, each edge device should run a small language model that matches its computational capacity. For example, while the central model may have 100 billion parameters, a resource-constrained mobile device might only handle a 100-million parameter model, representing a 1000x size difference. The key challenge then becomes how to effectively aggregate and fuse knowledge from these much smaller models into the large target model. We need novel techniques that can meaningfully combine insights from models operating at radically different scales while preserving the unique contributions of each small model. This requires fundamentally rethinking traditional model fusion approaches [96, 97] to handle such extreme parameter count disparities.

5.2 Heterogeneous Device Compute Sharing: from Node to Cluster

The second direction is to enable efficient compute resource sharing across heterogeneous devices by treating them as a unified compute cluster rather than independent nodes. Consider a smart home environment where multiple devices—smartphones, laptops, and desktop computers—could form a collaborative compute cluster. While each individual device has limited resources, their collective computing power could be substantial. For example, a laptop could handle intensive computational tasks, smartphones could manage coordination and lightweight processing, and desktop computers could contribute their onboard computing power. Meanwhile, other IoT devices such as smart speakers, security cameras, and vehicles could serve as data sources, providing valuable real-world inputs like voice commands, visual feeds, and environmental parameters. The language model would effectively run and train across this entire device cluster, leveraging both computing power and diverse training data from the environment. This distributed execution requires new frameworks that can intelligently decompose and distribute model computations based on each device’s capabilities and current load. The system must dynamically balance workloads - when the security cameras are idle at night, they could take on additional compute tasks, while during peak usage hours, the load could shift to other devices. This requires innovations in real-time resource allocation, task scheduling across heterogeneous hardware, and efficient inter-device communication protocols to ensure the collective computing power is optimally utilized [98].

6 Conclusion

In this position paper, we have argued that leveraging massive distributed edge devices can break barriers of data and computing wall, and everyone can participate in training large models with small edge devices. Our comprehensive analysis demonstrated the vast untapped potential of edge resources, with smartphone data volume reaching approximately 33.1 EB and a combined computing power of around 9278 EFLOPS in the past 5 years. These edge resources offer unique advantages in terms of data diversity, privacy, real-time context, and computing efficiency. This paradigm shift towards distributed training could democratize AI development and open an exciting new chapter in the scaling of foundation models.

References

- [1] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- [2] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- [4] OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [5] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.
- [6] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.
- [7] David Patterson, Joseph Gonzalez, Quoc V Le, Chen Liang, Lluís-Miquel Munguia, Daniel Rothchild, David So, Maud Texier, and Jeff Dean. Carbon emissions and large neural network training. *arXiv preprint arXiv:2104.10350*, 2021.
- [8] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [9] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255. IEEE, 2009.
- [10] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- [11] Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- [12] Meta. Introducing llama 3.1: Our most capable models to date. <https://ai.meta.com/blog/meta-llama-3-1/>, 2024. Accessed: 2025-01-22.
- [13] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- [14] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21:1–67, 2020.
- [15] DeepLearning.AI. Introduction to federated learning. <https://www.deeplearning.ai/short-courses/intro-to-federated-learning/>, 2024. Accessed: 2025-02-23.
- [16] Katherine Lee, Daphne Ippolito, Andrew Nystrom, Chiyuan Zhang, Douglas Eck, Chris Callison-Burch, and Nicholas Carlini. Deduplicating training data makes language models better. *arXiv preprint arXiv:2107.06499*, 2021.
- [17] Stella Biderman, Kieran Schoelkopf, Anthony Weiss, and David Noever. Data governance in the age of large language models. *arXiv preprint arXiv:2211.09911*, 2022.

- [18] Pablo Villalobos, Anson Ho, Jaime Sevilla, Tamay Besiroglu, Lennart Heim, and Marius Hobbhahn. Position: Will we run out of data? limits of llm scaling based on human-generated data. In *International Conference on Machine Learning*, pages 49523–49544. PMLR, 2024.
- [19] Hao Chen, Abdul Waheed, Xiang Li, Yidong Wang, Jindong Wang, Bhiksha Raj, and Marah I Abdin. On the diversity of synthetic data and its impact on training large language models. *arXiv preprint arXiv:2410.15226*, 2024.
- [20] Emily Wenger. Ai produces gibberish when trained on too much ai-generated data, 2024.
- [21] Xiao Fang, Shangkun Che, Minjia Mao, Hongzhe Zhang, Ming Zhao, and Xiaohang Zhao. Bias of ai-generated content: an examination of news produced by large language models. *Scientific Reports*, 14(1):5224, 2024.
- [22] Protection Regulation. General data protection regulation. *Intouch*, 25:1–5, 2018.
- [23] Dean Hardy-White. Are ai scaling laws hitting a wall? <https://www.linkedin.com/pulse/ai-scaling-laws-hitting-wall-dean-hardy-white-xchfe/>, 2024. Accessed: 2025-01-22.
- [24] xAI. Introducing grok-3. <https://x.ai/blog/grok-3>, 2025. Accessed: 2025-02-23.
- [25] Neil C Thompson, Kristjan Greenewald, Keeheon Lee, and Gabriel F Manso. Deep learning’s diminishing returns. *IEEE Spectrum*, 58(10):50–55, 2021.
- [26] Or Sharir, Barak Peleg, and Yoav Shoham. The cost of training nlp models: A concise overview. *arXiv preprint arXiv:2004.08900*, 2020.
- [27] Roy Schwartz, Jesse Dodge, Noah A Smith, and Oren Etzioni. Green ai. *Communications of the ACM*, 63(12):54–63, 2020.
- [28] Andrei Hagiu and Julian Wright. Artificial intelligence and competition policy. *International Journal of Industrial Organization*, page 103134, 2025.
- [29] Jack Thompson, Amanda Askill, and Jeffrey Song. Frontier ai regulation: Managing emerging risks to public safety. *arXiv preprint arXiv:2207.05257*, 2022.
- [30] Pablo Villalobos and Anson Ho. Trends in training dataset sizes. *Epoch AI Blog*, 2022.
- [31] Pablo Villalobos, Anson Ho, Jaime Sevilla, Tamay Besiroglu, Lennart Heim, and Marius Hobbhahn. Will we run out of data? limits of llm scaling based on human-generated data. *arXiv preprint arXiv:2211.04325*, pages 13–29, 2024.
- [32] Jaime Sevilla, Lennart Heim, Anson Ho, Tamay Besiroglu, Marius Hobbhahn, and Pablo Villalobos. Compute trends across three eras of machine learning. In *2022 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2022.
- [33] Bingbin Liu, Sébastien Bubeck, Ronen Eldan, Janardhan Kulkarni, Yuanzhi Li, Anh Nguyen, Rachel Ward, and Yi Zhang. Tinygsm: Achieving 80% on gsm8k with small models. *arXiv preprint arXiv:2312.09237*, 2023.
- [34] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget. *arXiv preprint arXiv:2305.17493*, 2023.
- [35] Elvis Dohmatob, Yunzhen Feng, Arjun Subramonian, and Julia Kempe. Strong model collapse. *arXiv preprint arXiv:2410.04840*, 2024.
- [36] Sina Alemohammad, Jose Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoobi, and Richard G Baraniuk. Self-consuming generative models go mad. *arXiv preprint arXiv:2307.01850*, 2023.
- [37] Li Fan, Kaiping Chen, Dilip Krishnan, Dina Katabi, Phillip Isola, and Yuandong Tian. Scaling laws of synthetic images for model training. *arXiv preprint arXiv:2306.09387*, 2023.

- [38] Marius Hobbhahn, Lennart Heim, and Gökçe Aydos. Trends in machine learning hardware, 2023. Accessed: 2025-01-27.
- [39] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [40] Henry Kressel. The end of moore’s law? innovation in computer systems continues. *Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research*, 2023.
- [41] Benzinga Staff. Apple, nvidia secure future with taiwan semi’s advanced chips as ai demand soars. *Benzinga*, June 2024.
- [42] ScaleFlux Research. Ai’s hardware hunger: The global semiconductor supply chain under pressure. *ScaleFlux Insights*, 2024. Accessed: 2025-01-27.
- [43] Statista global data volume. Volume of data/information created, captured, copied, and consumed worldwide from 2010 to 2025, 2023.
- [44] Statista IoT device data volume. Internet of things (iot) connected devices data size worldwide from 2019 to 2025, 2023.
- [45] Grand View Research. Edge computing market size & share analysis report, 2023-2030, 2023.
- [46] BankMyCell. How many smartphones are in the world?, 2023.
- [47] Seagate. Dataage white paper: The digitization of the world – from edge to core, 2019.
- [48] Seagate. Rethink data report 2020, 2020.
- [49] Sabuzima Nayak, Ripon Patgiri, Lilapati Waikhom, and Arif Ahmed. A review on edge analytics: Issues, challenges, opportunities, promises, future directions, and applications. *Digital Communications and Networks*, 10(3):783–804, 2024.
- [50] Cavli Wireless. Edge Computing for IoT, Real-Time Data and Low Latency Processing, 2023. Accessed:2025-01-22.
- [51] Shiwen Ni, Haihong Wu, Di Yang, Qiang Qu, Hamid Alinejad-Rokny, and Min Yang. Small language model as data prospector for large language model. *arXiv preprint arXiv:2412.09990*, 2024.
- [52] Apple Inc. iphone 16 pro and 16 pro max - technical specifications, 2024.
- [53] NVIDIA. Nvidia jetson agx orin tflops specifications, 2023. Forum discussion clarifying sparse vs. dense TFLOPS.
- [54] NanoReview.net. NanoReview.net - Gadget Specifications and Comparisons. <https://nanoreview.net>, 2025. Accessed: 2025-02-23.
- [55] Canalys. Canalys Newsroom - Market Analysis and Research. <https://canalys.com/newsroom>, 2025. Accessed: 2025-02-23.
- [56] Zhenyan Lu, Xiang Li, Dongqi Cai, Rongjie Yi, Fangming Liu, Xiwen Zhang, Nicholas D Lane, and Mengwei Xu. Small language models: Survey, measurements, and insights. *arXiv preprint arXiv:2409.15790*, 2024.
- [57] Fali Wang, Zhiwei Zhang, Xianren Zhang, Zongyu Wu, Tzuhao Mo, Qiuhao Lu, Wanjing Wang, Rui Li, Junjie Xu, Xianfeng Tang, et al. A comprehensive survey of small language models in the era of large language models: Techniques, enhancements, applications, collaboration with llms, and trustworthiness. *arXiv preprint arXiv:2411.03350*, 2024.
- [58] Chien Van Nguyen, Xuan Shen, Ryan Aponte, Yu Xia, Samyadeep Basu, Zhengmian Hu, Jian Chen, Mihir Parmar, Sasidhar Kunapuli, Joe Barrow, et al. A survey of small language models. *arXiv preprint arXiv:2410.20011*, 2024.

- [59] Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. Tinybert: Distilling bert for natural language understanding. *arXiv preprint arXiv:1909.10351*, 2020.
- [60] Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. Albert: A lite bert for self-supervised learning of language representations. *arXiv preprint arXiv:1909.11942*, 2020.
- [61] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.
- [62] Paolo Glorioso, Quentin Anthony, Yury Tokpanov, Anna Golubeva, Vasudev Shyam, James Whittington, Jonathan Pilault, and Beren Millidge. The zamba2 suite: Technical report. *arXiv preprint arXiv:2411.15242*, 2024.
- [63] Xin Dong, Yonggan Fu, Shizhe Diao, Wonmin Byeon, Zijia Chen, Ameya Sunil Mahabaleshwar, Shih-Yang Liu, Matthijs Van Keirsbilck Bilicki, Ziyang Ma, Qingyao Ai, et al. Hymba: A hybrid-head architecture for small language models. *arXiv preprint arXiv:2411.13676*, 2024.
- [64] Maximilian Beck, Korbinian Pöppel, Markus Spanring, Andreas Auer, Oleksandra Prudnikova, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. xlstm: Extended long short-term memory. *arXiv preprint arXiv:2405.04517*, 2024.
- [65] Daria Soboleva, Faisal Al-Khateeb, Robert Myers, Jacob R Steeves, Joel Hestness, and Nolan Dey. SlimPajama: A 627B token cleaned and deduplicated version of RedPajama. <https://www.cerebras.net/blog/slimpajama-a-627b-token-cleaned-and-deduplicated-version-of-redpajama>, 2023.
- [66] Bo Peng, Eric Alcaide, Quentin Anthony, Alon Albalak, Pedro Arcadinho, Eric Cao, Xin Cui, Zihang Dai, Jeff Eissman, Orhan Firat, Sophia Fu, Cong Gao, Yanping Hu, Maarten Hughes, James Kenealy, Maxim Krikun, Sneha Li, Yanping Li, Xiang Liu, Lianmin Luo, David McAllester, Matthew Olson, Alec Patel, Reiner Pope, Noam Rao, Alex Roberts, Noam Shazeer, Aditya Siddhant, Yi Tay, Duy Tran, Jason Wang, and Wang Wei. RwkV: Reinventing rNNS for the transformer era. *arXiv preprint arXiv:2305.13048*, 2023.
- [67] Ian Magnusson, Akshita Bhagia, Valentin Hofmann, Luca Soldaini, Ananya Harsh Jha, Oyvind Tafjord, Dustin Schwenk, Evan Walsh, Yanai Elazar, Kyle Lo, et al. Paloma: A benchmark for evaluating language model fit. *Advances in Neural Information Processing Systems*, 37:64338–64376, 2024.
- [68] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2019.
- [69] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.
- [70] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.
- [71] Yao Fu, Hao Peng, Ashish Khotilovich, Liang Chen, and Yan Yang. Specializing smaller language models towards multi-step reasoning. *arXiv preprint arXiv:2301.12726*, 2023.
- [72] Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*, 2023.
- [73] Exo Labs. Exo: Run your own ai cluster at home with everyday devices. <https://github.com/exo-explore/exo>, 2025. Accessed: 2025-01-29.

- [74] Yiping Kang, Johann Hauswald, Cao Gao, Andrew M. Rovinski, Trevor Mudge, Jason Mars, and Lingjia Tang. Neurosurgeon: Collaborative intelligence between the cloud and mobile edge. In *Proceedings of the Twenty-Second International Conference on Architectural Support for Programming Languages and Operating Systems*, pages 615–629. ACM, 2017.
- [75] Lyudong Jin, Yanning Zhang, Yanhan Li, Shurong Wang, Howard H. Yang, Jian Wu, and Meng Zhang. Moe²: Optimizing collaborative inference for edge large language models. *arXiv preprint arXiv:2501.09410*, 2025. Submitted to IEEE/ACM Transactions on Networking.
- [76] En Li, Zhi Zhou, and Xu Chen. Edge intelligence: On-demand deep learning model co-inference with device-edge synergy. In *Proceedings of the 2018 ACM/IEEE Symposium on Edge Computing*, pages 31–46. IEEE, 2018.
- [77] Shengyuan Ye, Jiangsu Du, Liekang Zeng, Wenzhong Ou, Xiaowen Chu, Yutong Lu, and Xu Chen. Galaxy: A resource-efficient collaborative edge ai system for in-situ transformer inference. *arXiv preprint arXiv:2405.17245*, 2024.
- [78] Ji Lin, Ligeng Zhu, Wei-Ming Chen, Wei-Chen Wang, Chuang Gan, and Song Han. On-device training under 256kb memory. *Advances in Neural Information Processing Systems*, 35:22941–22954, 2022.
- [79] Han Cai, Chuang Gan, Tianzhe Wang, Zhekai Zhang, and Song Han. Tinytl: Reduce memory, not parameters for efficient on-device learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 11285–11297, 2020.
- [80] Xinchu Qiu, Javier Fernandez-Marques, Pedro PB Gusmao, Yan Gao, Titouan Parcollet, and Nicholas Donald Lane. Zerofl: Efficient on-device training for federated learning with local sparsity. In *International Conference on Learning Representations*, 2022.
- [81] Keisuke Sugiura and Hiroki Matsutani. Elasticzo: A memory-efficient on-device learning with combined zeroth- and first-order optimization. *arXiv preprint arXiv:2501.04287*, 2025.
- [82] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. Communication-efficient learning of deep networks from decentralized data. *Artificial intelligence and statistics*, pages 1273–1282, 2017.
- [83] Jiamu Bai, Daoyuan Chen, Bingchen Qian, Liuyi Yao, and Yaliang Li. Federated fine-tuning of large language models under heterogeneous language tasks and client resources. *arXiv e-prints*, pages arXiv–2402, 2024.
- [84] Yiyuan Yang, Guodong Long, Tianyi Zhou, Qinghua Lu, Shanshan Ye, and Jing Jiang. Federated adapter on foundation models: An out-of-distribution approach. *arXiv preprint arXiv:2505.01075*, 2025.
- [85] Zhuo Zhang, Yuanhang Yang, Yong Dai, Qifan Wang, Yue Yu, Lizhen Qu, and Zenglin Xu. Fedpetuning: When federated learning meets the parameter-efficient tuning methods of pre-trained language models. In *Annual Meeting of the Association of Computational Linguistics 2023*, pages 9963–9977. Association for Computational Linguistics (ACL), 2023.
- [86] Haokun Chen, Yao Zhang, Denis Krompass, Jindong Gu, and Volker Tresp. Feddat: an approach for foundation model finetuning in multi-modal heterogeneous federated learning. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence*, pages 11285–11293, 2024.
- [87] Jiangui Chen, Ruqing Zhang, Jiafeng Guo, Yixing Fan, and Xueqi Cheng. Fedmatch: Federated learning over heterogeneous question answering data. In *Proceedings of the 30th ACM international conference on information & knowledge management*, pages 181–190, 2021.
- [88] Flowerlab. Flower: A friendly federated ai framework, 2025.
- [89] Tao Fan, Yan Kang, Guoqiang Ma, Weijing Chen, Wenbin Wei, Lixin Fan, and Qiang Yang. Fate-llm: An industrial grade federated learning framework for large language models. *arXiv preprint arXiv:2310.10049*, 2023.

- [90] PrimeIntellect-ai. Opendiloco: An open-source framework for globally distributed low-communication training, 2025.
- [91] Lorenzo Sani, Alex Jacob, Zeyu Cao, Royson Lee, Bill Marino, Yan Gao, Dongqi Cai, Zexi Li, Wanru Zhao, Xinchu Qiu, et al. Photon: Federated llm pre-training. *arXiv preprint arXiv:2411.02908*, 2024.
- [92] Elliot Bolton, Abhinav Venigalla, Michihiro Yasunaga, David Hall, Betty Xiong, Tony Lee, Roxana Daneshjou, Jonathan Frankle, Percy Liang, Michael Carbin, et al. Biomedlm: A 2.7 b parameter language model trained on biomedical text. *arXiv preprint arXiv:2403.18421*, 2024.
- [93] Peter Kairouz, H Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Keith Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, et al. Advances and open problems in federated learning. *Foundations and Trends® in Machine Learning*, 14(1–2):1–210, 2021.
- [94] Samuel Horvath, Stefanos Laskaridis, Mario Almeida, Ilias Leontiadis, Stylianos Venieris, and Nicholas Lane. Fjord: Fair and accurate federated learning under heterogeneous targets with ordered dropout. *Advances in Neural Information Processing Systems*, 34:12876–12889, 2021.
- [95] Samiul Alam, Luyang Liu, Ming Yan, and Mi Zhang. Fedrolex: Model-heterogeneous federated learning with rolling sub-model extraction. *Advances in neural information processing systems*, 35:29677–29690, 2022.
- [96] Boris Velasevic, Rohit Parasnis, Christopher G Brinton, and Navid Azizan. On the effects of data heterogeneity on the convergence rates of distributed linear system solvers. In *2023 62nd IEEE Conference on Decision and Control (CDC)*, pages 8394–8399. IEEE, 2023.
- [97] N. Azizan-Ruhi, F. Lahouti, A. S. Avestimehr, and B. Hassibi. Distributed solution of large-scale linear systems via accelerated projection-based consensus. *IEEE Transactions on Signal Processing*, 67(14):3806–3817, July 2019.
- [98] Ziyu Zhao, Leilei Gan, Guoyin Wang, Yuwei Hu, Tao Shen, Hongxia Yang, Kun Kuang, and Fei Wu. Retrieval-augmented mixture of lora experts for uploadable machine learning. *arXiv preprint arXiv:2406.16989*, 2024.
- [99] Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- [100] Ceilidh Welsh, Susana Román García, Gillian C Barnett, and Raj Jena. Democratising artificial intelligence in healthcare: community-driven approaches for ethical solutions. *Future Healthcare Journal*, 11(3):100165, 2024.
- [101] Jiangchao Yao, Shengyu Zhang, Yang Yao, Feng Wang, Jianxin Ma, Jianwei Zhang, Yunfei Chu, Luo Ji, Kunyang Jia, Tao Shen, et al. Edge-cloud polarization and collaboration: A comprehensive survey for ai. *IEEE Transactions on Knowledge and Data Engineering*, 35(7):6866–6886, 2022.
- [102] Haotian Dong, Jingyan Jiang, Rongwei Lu, Jiajun Luo, Jiajun Song, Bowen Li, Ying Shen, and Zhi Wang. Beyond a single ai cluster: A survey of decentralized llm training. *arXiv preprint arXiv:2503.11023*, 2025.
- [103] Fanlong Zeng, Wensheng Gan, Yongheng Wang, and Philip S Yu. Distributed training of large language models. In *2023 IEEE 29th International Conference on Parallel and Distributed Systems (ICPADS)*, pages 840–847. IEEE, 2023.
- [104] Zirui Song, Bin Yan, Yuhao Liu, Miao Fang, Mingzhe Li, Rui Yan, and Xiuying Chen. Injecting domain-specific knowledge into large language models: a comprehensive survey. *arXiv preprint arXiv:2502.10708*, 2025.
- [105] Ming Xu. Medicalgpt: Training medical gpt model. <https://github.com/shibing624/MedicalGPT>, 2023.

- [106] Yinheng Li, Shaofei Wang, Han Ding, and Hang Chen. Large language models in finance: A survey. In *Proceedings of the fourth ACM international conference on AI in finance*, pages 374–382, 2023.
- [107] Shunyu Yao, Qingqing Ke, Qiwei Wang, Kangtong Li, and Jie Hu. Lawyer gpt: A legal large language model with enhanced domain knowledge and reasoning capabilities. In *Proceedings of the 2024 3rd International Symposium on Robotics, Artificial Intelligence and Information Engineering*, pages 108–112, 2024.
- [108] DeepMind. Alphaevolve: A gemini-powered coding agent for designing advanced algorithms, 2025. Accessed: 2025-05-19.
- [109] Yuxin Yao et al. The pursuit of fairness in artificial intelligence models: A survey, 2024.
- [110] Tian Li, Maziar Sanjabi, Ahmad Beirami, and Virginia Smith. Fair resource allocation in federated learning. In *International Conference on Learning Representations*, 2020.
- [111] Hongyi Wang, Mikhail Yurochkin, Yuekai Sun, Dimitris Papailiopoulos, and Yasaman Khazaeni. Optimizing federated learning on non-IID data with reinforcement learning. *IEEE International Conference on Computer Communications*, pages 1698–1707, 2020.
- [112] Mehryar Mohri, Gary Sivek, and Ananda Theertha Suresh. Agnostic federated learning. In *International Conference on Machine Learning*, pages 4615–4625, 2019.
- [113] Jiawen Kang, Zehui Xiong, Dusit Niyato, Han Ye, and Dong In Kim. Incentive design for efficient federated learning in mobile networks: A contract theory approach. In *IEEE VTS Asia Pacific Wireless Communications Symposium*, pages 1–5, 2019.
- [114] Latif U. Khan, Shashi Raj Pandey, Nguyen H. Tran, Walid Saad, Zhu Han, Minh N. H. Nguyen, and Choong Seon Hong. Federated learning for edge networks: Resource optimization and incentive mechanism. *IEEE Communications Magazine*, 57(10):94–100, 2019.
- [115] Zhilin Wang, Qin Hu, Ruinian Li, Minghui Xu, and Zehui Xiong. Incentive mechanism design for joint resource allocation in blockchain-based federated learning. *IEEE Transactions on Parallel and Distributed Systems*, 34(5):1536–1547, 2023.
- [116] Qiang Yang, Yang Liu, Tianjian Chen, and Yongxin Tong. Federated machine learning: Concept and applications. *ACM Transactions on Intelligent Systems and Technology (TIST)*, 10(2):1–19, 2019.
- [117] Nafiseh Sharghivand, Farnaz Derakhshan, Lena Mashayekhy, and Leyli Mohammadkhanli. An edge computing matching framework with guaranteed quality of service. *IEEE Transactions on Cloud Computing*, 10(3):1557–1570, 2020.
- [118] Yuchen Yang et al. Environmental burden of united states data centers in the artificial intelligence era, 2024.
- [119] Xiaoyu Li et al. Carbon footprint reduction for sustainable data centers in real-time, 2024.
- [120] Alfonso Capozzoli and Giulio Primiceri. Cooling systems in data centers: State of art and emerging technologies. *Energy Procedia*, 83:484–493, 2015.
- [121] Xinchu Qiu, Titouan Parcollet, Daniel J. Beutel, Taner Topal, Akhil Mathur, and Nicholas D. Lane. Can federated learning save the planet? *arXiv preprint arXiv:2010.06537*, 2021.
- [122] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. *Artificial Intelligence and Statistics*, pages 1273–1282, 2017.
- [123] Ryan Smith. Nvidia announces jetson tx2: Parker comes to nvidia’s embedded system kit. *IEEE Hot Chips*, 29, 2017.
- [124] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks. *arXiv preprint arXiv:1812.06127*, 2018.

- [125] Alexandre Lacoste, Alexandra Luccioni, Victor Schmidt, and Thomas Dandres. Quantifying the carbon emissions of machine learning. *arXiv preprint arXiv:1910.09700*, 2019.
- [126] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, volume 25, pages 1097–1105, 2012.
- [127] Neil C Thompson, Kristjan Greenewald, Keeheon Lee, and Gabriel F Manso. The computational limits of deep learning. *arXiv preprint arXiv:2007.05558*, 10, 2020.
- [128] Frank Rosenblatt. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological review*, 65(6):386, 1958.
- [129] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [130] Rajat Raina, Anand Madhavan, and Andrew Y Ng. Large-scale deep unsupervised learning using graphics processors. In *Proceedings of the 26th annual international conference on machine learning*, pages 873–880, 2009.
- [131] Norman P Jouppi, Cliff Young, Nishant Patil, and David Patterson. In-datacenter performance analysis of a tensor processing unit. In *Proceedings of the 44th annual international symposium on computer architecture*, pages 1–12, 2017.
- [132] Yu Emma Wang, Gu-Yeon Wei, and David Brooks. Benchmarking tpu, gpu, and cpu platforms for deep learning. *arXiv preprint arXiv:1907.10701*, 2019.
- [133] Dario Amodei and Danny Hernandez. Ai and compute. *OpenAI Blog*, 2, 2018.
- [134] CounterPoint. How many smartphones are in the world?, 2021.
- [135] Rongjie Yi, Xiang Li, Weikai Xie, Zhenyan Lu, Chenghua Wang, Ao Zhou, Shangguang Wang, Xiwen Zhang, and Mengwei Xu. Phonelm: an efficient and capable small language model family through principled pre-training. *arXiv preprint arXiv:2411.05046*, 2024.
- [136] Jeffrey Li, Alex Fang, Georgios Smyrnis, Maor Ivgi, Matt Jordan, Samir Yitzhak Gadre, Hritik Bansal, Etash Guha, Sedrick Scott Keh, Kushal Arora, et al. Datacomp-lm: In search of the next generation of training sets for language models. *Advances in Neural Information Processing Systems*, 37:14200–14282, 2024.
- [137] Raymond Li, Daniel Choi, Jordi Chung, et al. Starcoder: May the source be with you! *arXiv preprint arXiv:2305.06161*, 2023.
- [138] Meta AI. Meta releases llama 3.2. <https://about.fb.com/news/2024/09/introducing-llama-3-2-1b-3b/>, 2024.
- [139] Jinze Yang, Shuai Wang, Shuohang Ma, Jianbo Zheng, et al. Qwen2: Technical report. *arXiv preprint arXiv:2404.05169*, 2024.
- [140] Jinze Bai, Shuai Wang, Fei Xiong, Zhenyu Hou, et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023.
- [141] Google Team. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.
- [142] Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, Colin Raffel, Leandro von Werra, and Thomas Wolf. Smollm2: When smol goes big – data-centric training of a small language model, 2025.
- [143] Loubna Ben Allal, Anton Lozhkov, Guilherme Penedo, Thomas Wolf, and Leandro von Werra. Smollm-corpus, 2024.

- [144] Pascal Pfeiffer, Philipp Singer, Yauhen Babakhin, Gabor Fodor, Nischay Dhankhar, and Sri Satish Ambati. H2o-danube3 technical report. *arXiv preprint arXiv:2407.09276*, 2024.
- [145] Edward Hu, Wangchunshu Huang, et al. Minicpm: Unveiling the potential of small language models. *arXiv preprint arXiv:2402.03216*, 2024.
- [146] AI2. Dolma: An open corpus of high-quality english text for language model pre-training. <https://huggingface.co/datasets/allenai/dolma>, 2023.
- [147] Xinrun Du, Zhouliang Yu, Songyang Gao, Ding Pan, Yuyang Cheng, Ziyang Ma, Ruibin Yuan, Xingwei Qu, Jiaheng Liu, Tianyu Zheng, et al. Chinese tiny llm: Pretraining a chinese-centric large language model. *arXiv preprint arXiv:2404.04167*, 2024.
- [148] Dirk Groeneveld, Iz Beltagy, Pete Davis, et al. Olmo: Accelerating the science of language models. *arXiv preprint arXiv:2402.00838*, 2024.
- [149] Peiyuan Zhang, Guangtao Chen, et al. Tinyllama: An open-source small language model. *arXiv preprint arXiv:2401.02385*, 2024.
- [150] Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, et al. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.
- [151] Mojan Javaheripi, Jacob Lobo, et al. Phi-2: The surprising power of small language models. *arXiv preprint arXiv:2312.12397*, 2023.
- [152] Suriya Gunasekar, Yi Zhang, et al. Textbooks are all you need. *arXiv preprint arXiv:2306.11644*, 2023.
- [153] Sachin Mehta, Mohammad Hossein Sekhavat, Qingqing Cao, Maxwell Horton, Yanzi Jin, Chenfan Sun, Iman Mirzadeh, Mahyar Najibi, Dmitry Belenko, Peter Zatloukal, et al. Openelm: An efficient language model family with open training and inference framework. *arXiv preprint arXiv:2404.14619*, 2024.
- [154] Guilherme Penedo, Anis Crnisanin, Ethan Shen, et al. The refinedweb dataset for falcon llm: Outperforming curated corpora with web data. *arXiv preprint arXiv:2306.01116*, 2023.
- [155] Leo Gao, Stella Biderman, Sid Black, et al. The pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- [156] Omkar Thawakar, Ashmal Vayani, Salman Khan, Hisham Cholakkal, Rao M Anwer, Michael Felsberg, Tim Baldwin, Eric P Xing, and Fahad Shahbaz Khan. Mobillama: Towards accurate and lightweight fully transparent gpt. *arXiv preprint arXiv:2402.16840*, 2024.
- [157] Zechun Liu, Changsheng Zhao, Forrest Iandola, Chen Lai, Yuandong Tian, Igor Fedorov, Yunyang Xiong, Ernie Chang, Yangyang Shi, Raghuraman Krishnamoorthi, et al. Mobilellm: Optimizing sub-billion parameter language models for on-device use cases. In *Forty-first International Conference on Machine Learning*, 2024.
- [158] Saurav Muralidharan, Sharath Turuvekere Sreenivas, Raviraj Joshi, Marcin Chochowski, Mostofa Patwary, Mohammad Shoeybi, Bryan Catanzaro, Jan Kautz, and Pavlo Molchanov. Compact language models via pruning and knowledge distillation. *Advances in Neural Information Processing Systems*, 37:41076–41102, 2024.
- [159] Arindam Mitra, Subhabrata Mukherjee, et al. Orca 2: Teaching small language models how to reason. *arXiv preprint arXiv:2311.11045*, 2023.
- [160] Subhabrata Mukherjee, Arindam Mitra, Ganesh Jawahar, Vicente Ordonez, and Kai-Wei Chang. Orca 2: Teaching small language models how to reason. *arXiv preprint arXiv:2312.02558*, 2023.
- [161] Shayne Longpre, Le Hou, Tu Vu, et al. The flan collection: Designing data and methods for effective instruction tuning. *arXiv preprint arXiv:2301.13688*, 2023.

- [162] Kehai Zhang, Zhuocheng Chen, et al. Towards making the most of chatgpt for machine translation. *arXiv preprint arXiv:2309.02654*, 2023.
- [163] Databricks. Free dolly: Introducing the world’s first truly open instruction-tuned llm. <https://www.databricks.com/blog/2023/04/12/dolly-first-open-commercially-viable-instruction-tuned-llm>, 2023.
- [164] Minghao Wu, Abdul Waheed, Chiyu Zhang, Muhammad Abdul-Mageed, and Alham Fikri Aji. Lamini-lm: A diverse herd of distilled models from large-scale instructions. *arXiv preprint arXiv:2304.14402*, 2023.
- [165] Elias Frantar, Saleh Ashkboos, Torsten Hoefer, et al. Sparsegpt: Massive language models can be accurately pruned in one-shot. *arXiv preprint arXiv:2301.00774*, 2023.
- [166] Zongyu Sun, Chen Chen, Zhitao Zhang, et al. A simple and effective pruning approach for large language models. *arXiv preprint arXiv:2306.11695*, 2023.
- [167] Mingyang Zhang, Hao Chen, Chunhua Shen, Zhen Yang, Linlin Ou, Xinyi Yu, and Bohan Zhuang. Loraprune: Structured pruning meets low-rank parameter-efficient fine-tuning. *arXiv preprint arXiv:2305.18403*, 2023.
- [168] Yu Men, Xingyu Zhang, Ruiqi Sun, et al. Shortgpt: Layers in large language models are more redundant than you expect. *arXiv preprint arXiv:2402.18952*, 2024.
- [169] Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Huaijie Wang, Lingxiao Ma, Fan Yang, Ruiping Wang, Yi Wu, and Furu Wei. Bitnet: Scaling 1-bit transformers for large language models. *arXiv preprint arXiv:2310.11453*, 2023.
- [170] Shuming Ma, Hongyu Zhao, Lingxiao Xue, et al. The era of 1-bit llms: All large language models are in 1.58 bits. *arXiv preprint arXiv:2402.17764*, 2024.
- [171] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. *arXiv preprint arXiv:2305.14314*, 2023.
- [172] Sehoon Kim, Coleman Hooper, Amir Gholami, et al. Squeezellm: Dense-and-sparse quantization. *arXiv preprint arXiv:2306.07629*, 2023.
- [173] Karan Goel. The on-device intelligence update, 2024.
- [174] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [175] Hyung Won Chung, Le Hou, et al. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*, 2022.
- [176] Together AI. Together ai: The ai acceleration cloud. <https://www.together.ai/>, 2023.
- [177] FLock. Flock: Federated machine learning on the blockchain. <https://www.flock.io/>, 2023.
- [178] alibaba. Federatedscope: An easy-to-use federated learning platform. <https://github.com/alibaba/FederatedScope>, 2024.
- [179] FedML-AI. Fedml: The unified and scalable ml library for large-scale distributed training, model serving, and federated learning. <https://github.com/FedML-AI/FedML>, 2024.
- [180] Rui Ye, Rui Ge, Xinyu Zhu, Jingyi Chai, Du Yaxin, Yang Liu, Yanfeng Wang, and Siheng Chen. Fedllm-bench: Realistic benchmarks for federated learning of large language models. *Advances in Neural Information Processing Systems*, 37:111106–111130, 2025.

A Impact Statements

The shift from centralized to distributed training of large models, may introduce new technical and societal challenges and have the potential to fundamentally reshape the AI landscape.

A.1 AI Monopoly and Democratization

The current AI landscape is characterized by significant concentration of power among a few tech giants, primarily due to their monopoly over massive computing resources and data centers [99]. This monopolistic trend has intensified with companies like OpenAI increasingly moving towards closed systems. While open-source alternatives like Llama [10], Deepseek [11] and other community-driven models have made strides towards democratization by releasing model parameters and technical reports [100], the gap in computational resources and data access between major AI companies and other players remains substantial and continues to widen. This disparity in resources allows tech giants to maintain their absolute dominance in determining the direction of AI development, raising concerns about AI democratization.

Edge device-based collaborative training presents a promising pathway to democratize AI development [101]. By leveraging the collective computing power of millions of edge devices, this approach could effectively challenge the existing monopolistic structure [102]. This democratization of AI training through edge devices could fundamentally reshape the structure of responsibilities and authorities. If everyone can participate in training LLMs, the AI landscape could fundamentally change. Training decisions would shift from companies to communities, creating shared responsibility for model development [103]. Global participation would help models reflect diverse cultural perspectives, while allowing communities to adapt models for their local needs. Furthermore, this decentralized approach could foster a more competitive and innovative AI ecosystem. When the barriers to entry for AI model training are lowered, we can expect to see a broader range of specialized models emerging (like [104–108]), better suited to local needs and diverse use cases.

A.2 Fairness and Incentive Mechanisms

The distributed training paradigm introduces new considerations for model fairness and bias mitigation [109, 110]. When training occurs across diverse edge devices, the resulting models can potentially better reflect the heterogeneous nature of user populations [111]. However, this approach also raises concerns about participation bias, where differences in device capabilities or user engagement could lead to underrepresentation of certain groups [93]. To address these challenges, researchers have proposed various fairness-aware federated learning algorithms [112] that aim to ensure equitable model performance across different demographic groups and device types [110].

To sustain a distributed training ecosystem, effective incentive mechanisms are crucial for motivating user participation [113]. Traditional approaches like computational resource sharing [114] and privacy-preserving reward systems have shown promise in encouraging user engagement. More innovative solutions include token-based reward systems [115] and reputation mechanisms [116] that compensate users for their contributions while maintaining system integrity. These incentive structures not only encourage consistent participation but also help ensure the quality of contributed training data [117], creating a sustainable ecosystem for collaborative AI development.

A.3 Carbon Footprint and Energy Efficiency

The shift from centralized to distributed training offers compelling environmental benefits [118]. Traditional data centers housing large language models face significant energy challenges [119] - their high-performance GPUs require extensive cooling systems that consume 30-40% of total energy [120]. In contrast, FL distributes computation across edge devices like smartphones and tablets that operate at much lower temperatures and power levels, eliminating industrial cooling needs [121].

FL also dramatically reduces data transmission energy costs. While centralized approaches require raw data transfer from millions of devices, FL only transmits lightweight model updates, substantially decreasing network energy overhead [122]. The hardware efficiency gap is striking - edge devices like the NVIDIA Tegra X2 consume just 7.5W during training compared to 250W for data center GPUs [123], translating to major carbon footprint reductions, particularly for simpler models [124]. By reducing reliance on power-hungry data centers and leveraging existing consumer devices, FL enables more sustainable AI development through optimized energy efficiency and minimized infrastructure needs. This combination of reduced cooling requirements, efficient hardware utilization,

and optimized data handling makes FL an environmentally responsible choice for the future of AI training [125]. As climate impact becomes increasingly critical, FL’s sustainability advantages position it as a key technology for green AI development.

B Historical Development and Current Challenges

B.1 Data: the fuel of LLMs

Early data-driven AI development As LLMs continue to achieve unprecedented success in artificial intelligence, understanding the role of data becomes increasingly crucial. From the early days of simple datasets to the modern era of massive data collections, data has consistently served as the lifeblood of AI, determining the upper bounds of model capabilities. The evolution of AI—marked by breakthroughs in computer vision, natural language processing, and beyond—can be traced back to the continuous expansion and refinement of data resources.

In the early stages of AI, despite relatively small data scales, the importance of data was already evident. The MNIST dataset, for instance, serves as a notable example. With 60,000 training images and 10,000 test images, it provided a crucial foundation for neural network research, demonstrating the fundamental role of data in model training [8]. As data scales expanded, the capabilities of deep learning models saw significant improvements. The emergence of ImageNet, which contains 14 million images across 21,000 synsets, revolutionized computer vision. This enabled deep learning models like AlexNet to learn complex visual features and achieve breakthrough progress in image recognition tasks, reducing error rates from 26.2% to 15.3% in the ILSVRC-2012 competition [9, 126]. ImageNet’s success stemmed not only from its scale but also from its high quality and diversity, laying the groundwork for subsequent large-scale data applications.

Era of massive data With the proliferation of the internet and advances in computing power, data scales have expanded dramatically, ushering AI into an era of massive data. GPT-3, for instance, was trained on 450 billion tokens, with a carefully curated mix of data sources: Common Crawl (60%), books (16%), Wikipedia (3%), and other internet-based text (21%) [5]. This massive dataset enabled GPT-3 to excel across various tasks, demonstrating the decisive role of data scale in model capabilities. Compared to early datasets like MNIST and ImageNet, GPT-3’s data scale and quality reached unprecedented heights, not only advancing natural language processing but also opening new possibilities for AI generalization.

Quality and diversity matter Beyond scale, data quality and diversity are crucial factors in model performance. ImageNet ensures data quality through rigorous validation, with each image verified by an average of 3.3 annotators and achieving 95% accuracy in its labels [9]. This precise annotation enables models to learn accurate visual features and excel in image classification tasks. In the realm of large language models, GPT-3’s training data underwent stringent cleaning and filtering, including deduplication, quality scoring based on document length and linguistic complexity, and content filtering for inappropriate content [5]. This high-quality data enables GPT-3 to generate coherent and accurate text. Furthermore, diversity is essential: ImageNet covers 1,000 object categories across various domains, while GPT-3’s training data spans multiple languages, genres, and knowledge domains, providing rich linguistic knowledge and contextual understanding.

Data as the ceiling for model capabilities A model’s capability depends on the knowledge it extracts from data, following empirically observed scaling laws. While increasing model parameters can enhance expressive power, without sufficient data, models cannot effectively utilize these parameters. DeepMind’s research on the Chinchilla model demonstrated that under the same compute budget, a 70B parameter model trained on 1.4T tokens outperforms a 280B parameter model trained on 0.35T tokens, achieving a 30% reduction in loss while using the same compute resources [2]. This finding directly supports the notion that data acts as a ceiling for model capabilities. Additionally, Meta’s research shows that while Llama 2 (70B) has 70 billion parameters, its performance largely benefits from training on 2T tokens of high-quality data, with particular emphasis on academic papers, code repositories, and books that enhance its reasoning capabilities [10]. These studies emphasize data’s central role in model training and suggest that optimal model scaling requires a balanced increase in both parameters and training data.

Looking ahead From MNIST to ImageNet to GPT-3, advances in data scale, quality, and diversity have directly driven AI breakthroughs. Data remains the foundation of AI development, determining the upper limits of model capabilities. As we push the boundaries of LLM performance, the challenge of acquiring sufficient high-quality, diverse data becomes increasingly acute. Traditional data sources like the internet are showing signs of exhaustion, and concerns about data privacy and ownership are growing. This motivates the exploration of novel data acquisition approaches, such as leveraging edge devices and distributed data collection, which we will explore in subsequent sections. The future of LLMs may depend not just on scaling existing data sources, but on fundamentally rethinking how we collect, curate, and utilize data in AI training.

B.2 Computing power: the engine of LLMs

Early neural networks and CPU era Since the inception of neural networks, every breakthrough in the field of AI has been driven by the continuous improvement of computational power [127]. From the early multilayer perceptron (MLP) to the widely used large language models (LLM) today, the progress in computing power has always been a key engine for advancing AI.

As the prototype of neural networks, the MLP was initially used to solve linearly separable problems [128]. Due to its relatively low computational demand, it could run on traditional CPU environments. However, as the complexity of neural network models increased and application scenarios expanded, computational requirements gradually rose. The emergence of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) marked a surge in computational demands. CNN, through convolutional operations, effectively reduced the number of parameters, enhancing the computational efficiency of image processing tasks. Classic models such as LeNet [129] and AlexNet [126] achieved significant results in image classification, but this also led to a surge in computational resource demands. For example, AlexNet’s victory in the 2012 ImageNet competition was made possible by using the NVIDIA GTX 580 GPU, which significantly boosted computational performance [126].

GPU and TPU revolution With the growing scale of neural network models, GPUs gradually became indispensable computing tools [130]. The parallel computing capabilities of GPUs greatly accelerated the training process of neural networks, particularly in the field of deep learning. Meanwhile, specialized hardware for deep learning, such as Tensor Processing Units (TPUs), emerged [131]. Compared to GPUs, TPUs offer higher efficiency and lower power consumption when performing matrix operations and deep learning tasks [132], making them the preferred hardware for training large-scale neural networks.

Transformer era and computational demands As computational resources continued to expand, the scale of neural network model training also grew. The introduction of the Transformer architecture [39] revolutionized the field of natural language processing (NLP), especially with the launch of models like BERT [3] and the GPT series [5, 4], which pushed NLP technology to new heights. However, the self-attention mechanism in the Transformer architecture has a computational complexity of $O(n^2)$, where n represents the sequence length [39]. This means that as the model scale and sequence length increase, the required computational power grows exponentially. For example, training large language models like GPT-3 [5] and GPT-4 [4] involves trillions of parameters and requires thousands of GPUs or TPU nodes to support the process. This immense computational demand not only places extremely high requirements on hardware, but also on computational frameworks, storage, and communication bandwidth, creating unprecedented challenges [7].

Computing power as the key driver Every leap in Artificial Intelligence has been driven by computational power [133]. From multilayer perceptrons to convolutional neural networks, and the introduction of the Transformer architecture, every innovation in models has been accompanied by an explosive growth in computational needs [127]. Particularly in the era of large language models, computational power is not only the foundational tool for model training but also the core driving force behind breakthroughs in AI performance [1]. The success of large-scale models like GPT-4 validates that AI progress almost entirely depends on the support of more powerful computational resources [2].

C Smartphone Data Volume Estimation

In the absence of publicly available, granular data on per-user smartphone data generation patterns, we adopt a conservative estimation approach to approximate the total annual smartphone data

volume. While this method necessarily involves simplifications, it provides a robust lower-bound approximation that is sufficient to support our core arguments without compromising the validity of our conclusions.

Data volume estimation per smartphone: Based on industry reports [134], the average smartphone storage capacity reached 100 GB in 2020. To ensure a conservative estimate, we assume that only 1% of this storage capacity (equivalent to approximately 1 GB per smartphone) is actively used for data generation and storage, including local images, video information, and other types of user-generated content. This assumption aligns with baseline usage scenarios while intentionally underestimating actual data utilization.

Number of smartphones: The growth of the number of smartphone users is an important basis for estimating the total amount of data. For this, we have referred to data from market research institutions [46], which includes trends in changes to the number of smartphone users over time.

Based on the above statistical data, the total annual smartphone data volume D_{total} is calculated using the following formula:

$$D_{\text{total}}(\text{EB}) = N_{\text{users}} \times 1 \text{ GB/user} \times 10^{-3} \text{ (conversion from GB to EB)}, \quad (1)$$

where N_{users} represents the global smartphone user base in billions.

Substituting $N_{\text{users}} = 8.0 \times 10^9$ (representing 8 billion users) into Equation (1):

$$D_{\text{total}} = 8.0 \text{ GB/user} \times 10^{-3} = 8.0 \text{ EB}.$$

Our purpose is to establish a defensible lower bound for analysis. Even under these stringent assumptions, the derived volumes remain orders of magnitude higher than synthetic or centralized datasets, thereby reinforcing the strategic importance and value of edge-generated data. This conservative estimation underscores the critical need for scalable solutions capable of managing and leveraging such vast quantities of distributed data effectively.

Table 1: Trends in Smartphone Shipments and Compute Power. (Data source: [55]).

Company	Shipments (Million units)	Chip Performance Range (TFLOPS)	Total Compute Power Contribution (EFLOPS)
2020			
Samsung (20%)	255.5	1.20–1.53	349
Apple (16%)	207.2	0.65	135
Xiaomi (12%)	149.6	0.24–1.20	108
OPPO (9%)	119.4	0.24–1.20	86
vivo (9%)	112.6	0.24–1.20	81
Others (33%)	420.5	0.04–0.24	59
Overall: Shipments = 1265 Million, Compute Power = 817 EFLOPS			
2021			
Samsung (20%)	274.5	1.42–1.72	430
Apple (17%)	230.1	1.71–1.94	420
Xiaomi (14%)	191.2	0.82–1.74	240
OPPO (11%)	145.1	0.82–1.74	180
vivo (10%)	129.9	0.82–1.74	160
Others (28%)	379.4	0.27–0.82	207
Overall: Shipments = 1350 Million, Compute Power = 1637 EFLOPS			
2022			
Samsung (22%)	257.9	0.49–2.01	322
Apple (19%)	232.2	1.79	416
Xiaomi (13%)	152.7	1.01–3.49	351
OPPO (10%)	113.4	1.01–3.49	261
Transsion (6%)	73.1	0.24–0.98	44.6
Others (31%)	364.1	0.84–1.31	393
Overall: Shipments = 1193 Million, Compute Power = 1788 EFLOPS			
2023			
Apple (20%)	229.1	2.15	493
Samsung (20%)	225.5	2.01–2.77	539
Xiaomi (13%)	146.1	2.15–3.99	449
OPPO (9%)	100.7	2.15–3.99	309
Transsion (8%)	92.6	0.24–1.31	72
Others (30%)	347.9	0.24–2.15	416
Overall: Shipments = 1142 Million, Compute Power = 2278 EFLOPS			
2024			
Apple (18%)	225.9	1.91–2.29	474
Samsung (18%)	222.9	3.38–3.41	758
Xiaomi (14%)	168.6	3.38–4.95	703
Transsion (9%)	106.7	0.05–0.67	38
OPPO (8%)	103.6	3.38–4.95	432
Others (33%)	395.4	0.05–1.72	352
Overall: Shipments = 1223 Million, Compute Power = 2758 EFLOPS			

D Estimation of Smartphone Total Computational Power

To assess the (ideally) aggregate computational capabilities of smartphones globally, we estimate the total computing power, given the current lack of comprehensive statistical data in this domain. Our approach leverages two key data sources: the annual worldwide shipment volumes for major smartphone brands, and the computational performance specifications of mobile processors deployed in their devices during each corresponding year. The complete data underlying our analysis is presented in Table 1, which provides a detailed breakdown by manufacturer and time period. For quantitative analysis, we formulated a mathematical model to calculate the total computing power. Specifically, for any given year, we compute the aggregate computational capacity (C_{total}) by summing the contributions from each smartphone manufacturer (i). Each manufacturer’s contribution is determined by multiplying their total device shipments (N_i) by the average computing power of their mobile processors (P_i) for that year, expressed formally as:

$$C_{\text{total}} = \sum_i N_i \cdot P_i \quad (2)$$

This formulation enables us to systematically track the evolution of distributed computing power across the smartphone ecosystem while accounting for both market share dynamics and technological advancement in mobile processors. By maintaining conservative estimates for processor capabilities and focusing on verified shipment data, our analysis provides a reliable lower bound for the total computational resources available through smartphones.

E Small Language Model (SLM) Architectures and Training Methods

Table 2 presents a comprehensive overview of the Small Language Model (SLM) landscape, categorized by architectures and training methodologies, according to [57]. The table is organized into two main categories: (I) Transformer-Based Models, which represent the dominant architecture in current SLMs, and (II) Alternative Architecture Models, which explore novel approaches to achieve efficiency. The Transformer-Based section is further divided into models pre-trained from scratch, models derived from larger LLMs through knowledge distillation, and models created through various compression techniques (pruning, quantization, etc.). The Alternative Architecture section showcases emerging approaches like State Space Models (Mamba, Hymba), recurrent architectures (RWKV, xLSTM), and traditional encoder-decoder or encoder-only designs.

This classification showcases the architectural innovations and training methodologies that are driving the SLM field forward, providing essential technical foundations for deploying powerful AI capabilities on resource-constrained edge devices. By documenting various model sizes, training corpora, and development techniques, the table offers a comprehensive overview of cutting-edge approaches that enable sophisticated language processing directly on end-user devices. These advancements represent critical building blocks for the next generation of on-device AI systems that can operate efficiently without constant cloud connectivity while still delivering robust performance across diverse applications.

Table 2: Small Language Model (SLM) Architectures and Training Methods

Model	Sizes	Architecture	From Scratch	From LLMs	Training Method	Datasets
I. Transformer-Based Models						
<i>I.A. Pre-Trained from Scratch</i>						
PhoneLM [135]	0.5B; 1.5B	Transformer	✓		Pre-training	DCLM-baseline [136], StarCoderData [137]
Llama 3.2 [138]	1B; 3B	Transformer	✓		Pre-training, SFT, RLHF, DPO	Not released (9T tokens)
Qwen 1/1.5/2/2.5 [139, 140]	0.5B-7B	Transformer	✓		Pre-training	Not released
Gemma [141]	2B; 7B	Transformer	✓		Pre-training	Unknown
SmolLM2 [142]	135M-1.7B	Transformer	✓		Pre-training	SmolLM corpus [143]
H2O-Danube3 [144]	500M; 4B	Transformer	✓		Pre-training (multi-stage)	Unknown
MiniCPM [145]	1.2B; 2.4B	Transformer	✓		Pre-training	Dolma [146], C4 [14]
CT-LLM [147]	2B	Transformer	✓		Pre-training	MAP-CC
OLMo [148]	1B; 7B	Transformer	✓		Pre-training	Dolma [146] (multiple sources)
TinyLlama [149]	1B	Transformer	✓		Pre-training	SlimPajama [65]
Phi-series [150, 151]	1.3B-6.6B	Transformer	✓		Pre-training	CodeTextBook [152]
OpenELM [153]	270M-3B	Transformer	✓		Pre-training	RefinedWeb [154], PILE [155]
MobiLlama [156]	0.5B; 0.8B	Transformer	✓		Pre-training	LLM360 Amber
MobileLLM [157]	125M; 350M	Transformer	✓		Pre-training	Unknown (1T tokens)
<i>I.B. Derived from Larger Models</i>						
MINITRON [158]	4B	Transformer		✓	Distillation, Pruning	8T tokens from Nemotron-4
Orca/Orca 2 [159, 160]	7B; 13B	Transformer		✓	Distillation	Orca 2 dataset, FLAN-v2 [161]
MINIMA [162]	3B	Transformer		✓	Distillation (from Llama-2-7B)	Pile [155], Wudao
Dolly-v2 [163]	3B; 7B	Transformer		✓	Instruction tuning (from Pythia)	Databricks-dolly-15k
LaMini-LM [164]	61M-7B	Transformer		✓	Distillation	LaMini instruction dataset
<i>I.C. Model Compression Approaches</i>						
SparseGPT [165]	Various	Transformer		✓	Unstructured Pruning	Not applicable
Wanda [166]	Various	Transformer		✓	Unstructured Pruning	Not applicable
LoRAPrune [167]	Various	Transformer		✓	Unstructured Pruning	Not applicable
ShortGPT [168]	Various	Transformer		✓	Structured Pruning	Not applicable
BitNet/BitNet [169, 170]	b1.58	Transformer		✓	Quantization (QAT)	Not applicable
QLoRA [171]	Various	Transformer		✓	Quantization, Low-Rank	Various fine-tuning datasets
SqueezeLLM [172]	Various	Transformer		✓	Quantization (PTQ)	Not applicable
II. Alternative Architecture Models						
Mamba [61]	125M-1.3B	Mamba	✓		Pre-training	Pile [155]
Rene [173]	1.3B	Mamba	✓		Pre-training	Dolma-1.7 [146]
Zamba2 [62]	2.7B	Mamba	✓		Pre-training	Not specified
Hymba [63]	125M-1.5B	Hymba	✓		Pre-training	DCLM-Baseline [136]
xLSTM [64]	125M-1.3B	xLSTM	✓		Pre-training	SlimPajama [65]
RWKV [66]	169M-14B	RNN	✓		Pre-training	Pile [155]
Specialized FlanT5 [71]	250M-3B	Encoder-Decoder		✓	Instruction Tuning	GSM8K [174]
FlanT5 [175]	80M-3B	Encoder-Decoder		✓	Instruction Tuning	Muffin, T0-SF, SNI and CoT
T5 [14]	60M-3B	Encoder-Decoder	✓		Pre-training	C4 [14]
DistilBERT [69]	66M	Encoder-only		✓	Distillation (from BERT)	Wikipedia, BookCorpus
TinyBERT [59]	14.5M	Encoder-only		✓	Distillation (from BERT)	Wikipedia, BookCorpus
ALBERT [60]	12M-18M	Encoder-only	✓		Pre-training (parameter sharing)	Wikipedia, BookCorpus

F Distributed Collaborative Frameworks

Distributed collaborative frameworks enable the deployment, training, and fine-tuning of language models across multiple devices or servers. Table 3 presents a comparison of prominent frameworks in this domain. These frameworks can be broadly categorized into three types: cloud-based platforms that offer centralized resources for distributed computing, federated learning systems that enable training across decentralized data sources while preserving privacy, and fully decentralized frameworks that distribute computation across peer nodes. Some frameworks like Neurosurgeon [74], MoE² [75], Edgent [76], and Galaxy [77] focus on collaborative inference by partitioning models between edge devices and servers. Others, such as FlexLoRA [83], FedFM [84], FedPETuning [85], and Photon [91], specialize in federated fine-tuning of large language models while maintaining data privacy. These frameworks are essential for enabling efficient deployment of language models in

resource-constrained environments and for scenarios requiring privacy preservation or operation in disconnected settings.

Framework	Distributed Capabilities				Type	Privacy	License
	Inference	Training	Pretraining	Fine-tuning			
exo-explore/exo [73]	✓				Decentralized		MIT
Together AI [176]	✓		✓	✓	Cloud		Commercial
FLock Platform [177]		✓		✓	Federated, Blockchain	✓	Apache 2.0
OpenDiloco [90]		✓	✓		Decentralized		Apache 2.0
FederatedScope [178]		✓		✓	Federated	✓	Apache 2.0
FedML [179]		✓		✓	Federated	✓	Apache 2.0
Flower [88]		✓		✓	Federated	✓	Apache 2.0
FATE-LLM [89]		✓		✓	Federated	✓	Apache 2.0
FedLLM [180]		✓	✓	✓	Federated	✓	CC BY-NC 4.0

Table 3: Comparison of Distributed Machine Learning Frameworks